

Roof Lifecycle Cost Analysis

Laurel market
October 2, 2026

Planned maintenance is projected to save the owner of a typical Laurel medical facility **\$704K** over the next 20 years.

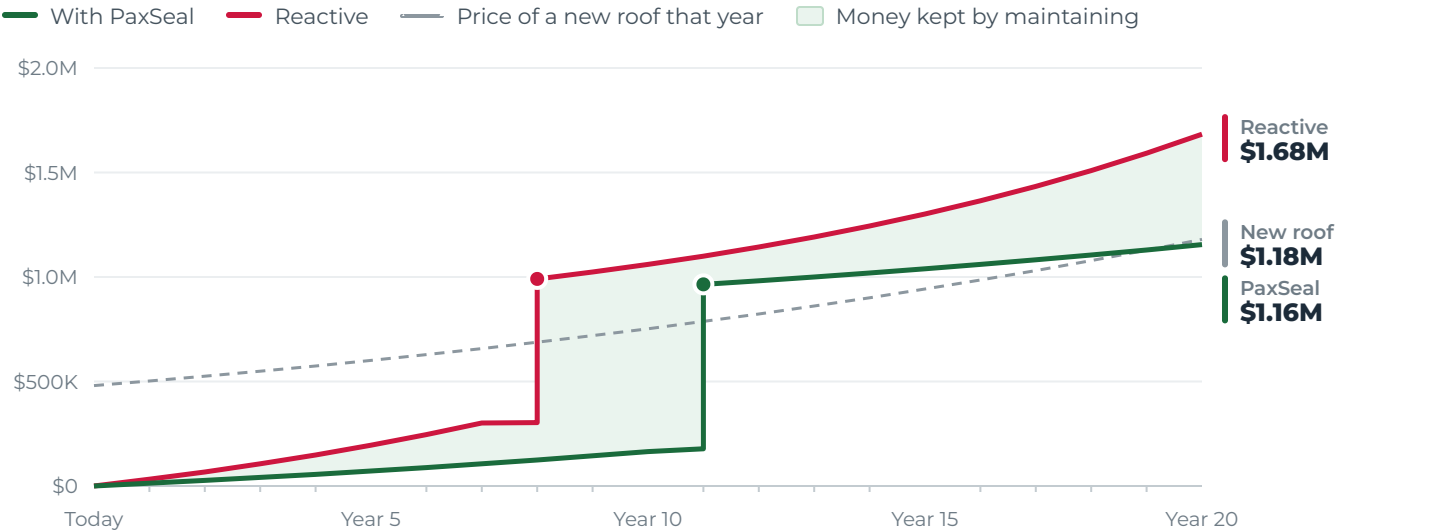
That is 36.9% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$357K to \$922K. In today's dollars the estimate is worth \$364K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Laurel	Medical	60,000 sq ft	Mod bit	8 yrs Good	\$480K

PAXSEAL NET COST \$1.2M \$1.16M spent plus roof wear, net of roof life left	REACTIVE NET COST \$1.91M \$1.68M spent plus roof wear, net of roof life left	EMERGENCY COSTS CUT \$859K Unbudgeted repairs and storm damage avoided	NEW ROOF PRICE 2× in 15 yrs \$480K today, \$1.18M by year 20
---	---	--	--

Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$231K	—	-\$231K
Repairs paid by owner	\$117K	\$966K	+\$849K
Expected storm damage	\$20K	\$30K	+\$10K
Roof replacements	\$787K	\$688K	-\$99K
Current roof value used	\$288K	\$288K	—
Credit for roof life left	-\$240K	-\$64K	+\$176K
Net 20-year cost	\$1.2M	\$1.91M	+\$704K

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 4.2× the cost in a medical facility, saving \$849K.
- 3 **Patient care keeps running.** A leak in a clinical area triggers infection-control work, closes rooms and can damage equipment.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$4.05 of other roof costs.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

~3.5 major regional roof-damaging events per 20 years (2010 snow, 2012 derecho, 2016 blizzard). Expected storm damage to this roof over 20 years is **\$20K** if maintained and **\$30K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	■■■■■
Strong wind	High	■■■■■
Tornado	High	■■■■■
Hurricane	Moderate	■■■■■
Winter weather	Very High	■■■■■
Ice storm	Moderate	■■■■■

FEMA National Risk Index rating

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$8.00/sq ft	Laurel contractor price ranges
Reactive repair cost	3× planned × 1.4	Industry estimates; no primary study found. ×1.4 for leak damage in a medical facility
Storm damage	0.13%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$8,600/yr	PAX program estimate for this roof size. Your proposal will state the actual fee

Estimates, not a quote. Projections from Laurel market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance.